4.1. ALTERNATING QUANTITY

Quantity which varies periodically with time is known as the alternating quantity. It may be voltage or current. Some waveforms of alternating quantities are shown in Fig. 4.1.

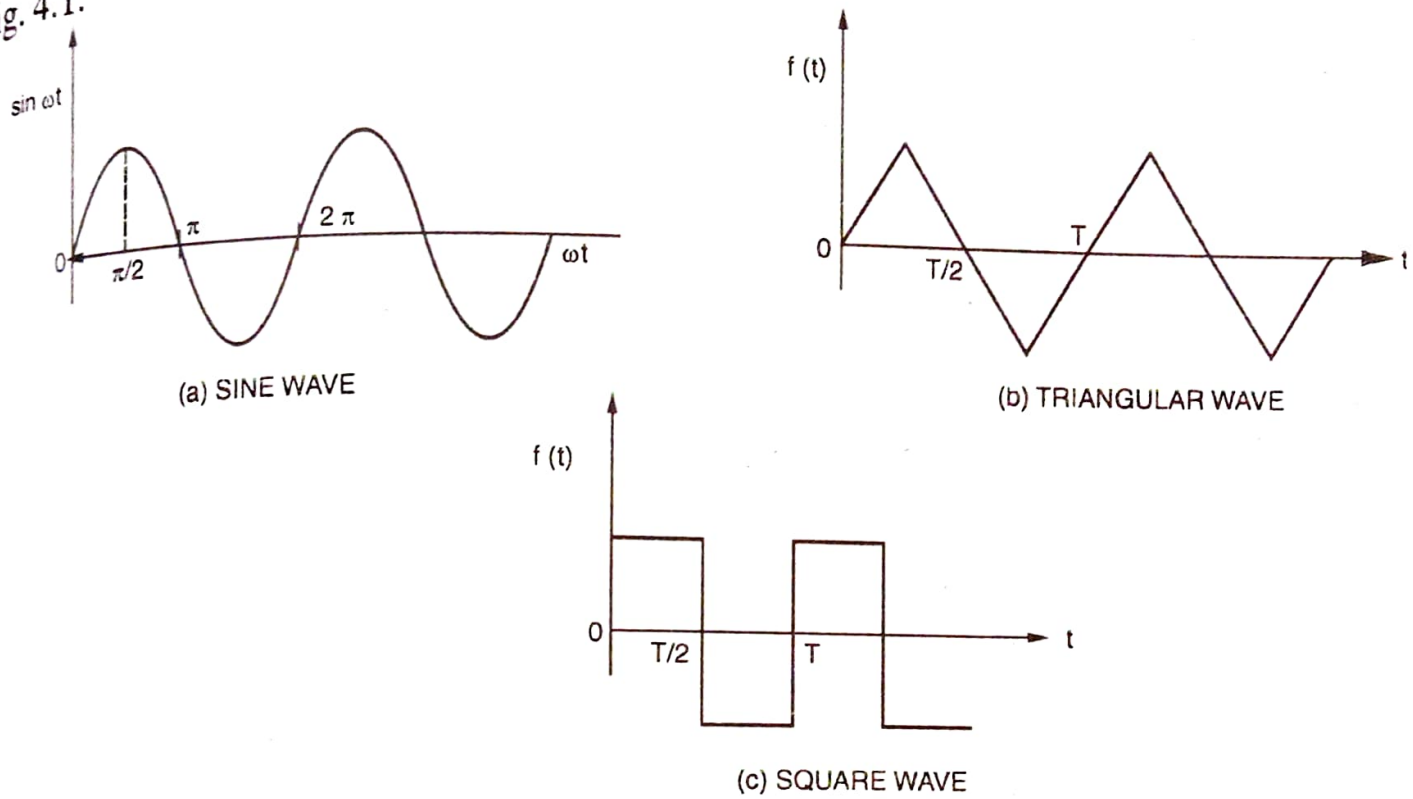


Fig. 4.1. Alternating waveform.

4.2. BASIC DEFINITIONS ABOUT ALTERNATING QUANTITY

Here we take the example of sinusoidal form of alternating current.

Cycle. It is the one complete set of positive and negative half of any alternating quantity.

Time period. It is the time required in second to complete the one cycle of any alternating quantity.

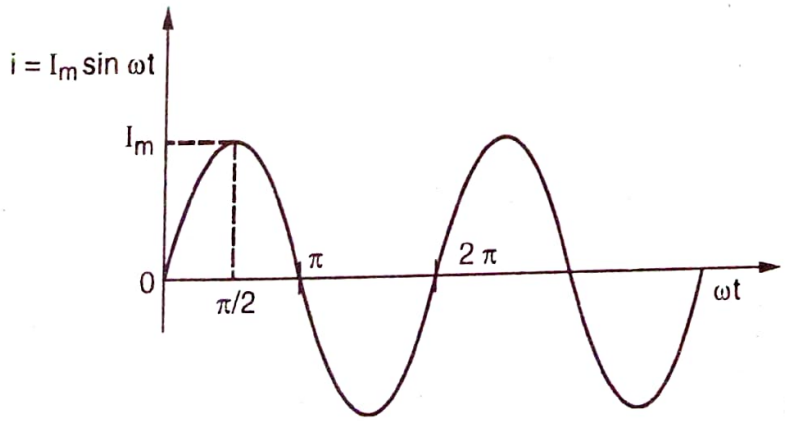


Fig. 4.2. Sinusoidal waveform.

Frequency. Number of cycles per second is known as the frequency of alternating quantity. It is the reciprocal of the time period and its unit is cycle per second or Hertz.

$$f = \frac{1}{T} \quad \text{in Hertz}$$

In India the frequency of alternating supply is 50 hertz, while in America and Japan it is 60 hertz.

Amplitude. It is the peak or maximum value of the alternating quantity. In Fig. 4.1, I_m denotes the amplitude of the current wave.

Equation of Sinusoidal alternating quantity. The sinusoidal alternating current is written as,

$$i = I_m \sin \omega t \qquad ...(4.1)$$

Similarly we can write for voltage,

$$v = V_m \sin \omega t \qquad ...(4.2)$$

where i, v = Instantaneous value of current and voltage,

I_m , V_m = Maximum values

$\omega = 2\pi f$ [f = frequency in Hz]

$$= \frac{2\pi}{T}$$

[T = Time period in sec] ...(4.3)



$\omega = 2\pi f$ [f = frequency in Hz]

$$= \frac{2\pi}{T}$$

[T = Time period in sec] ...(4.3)

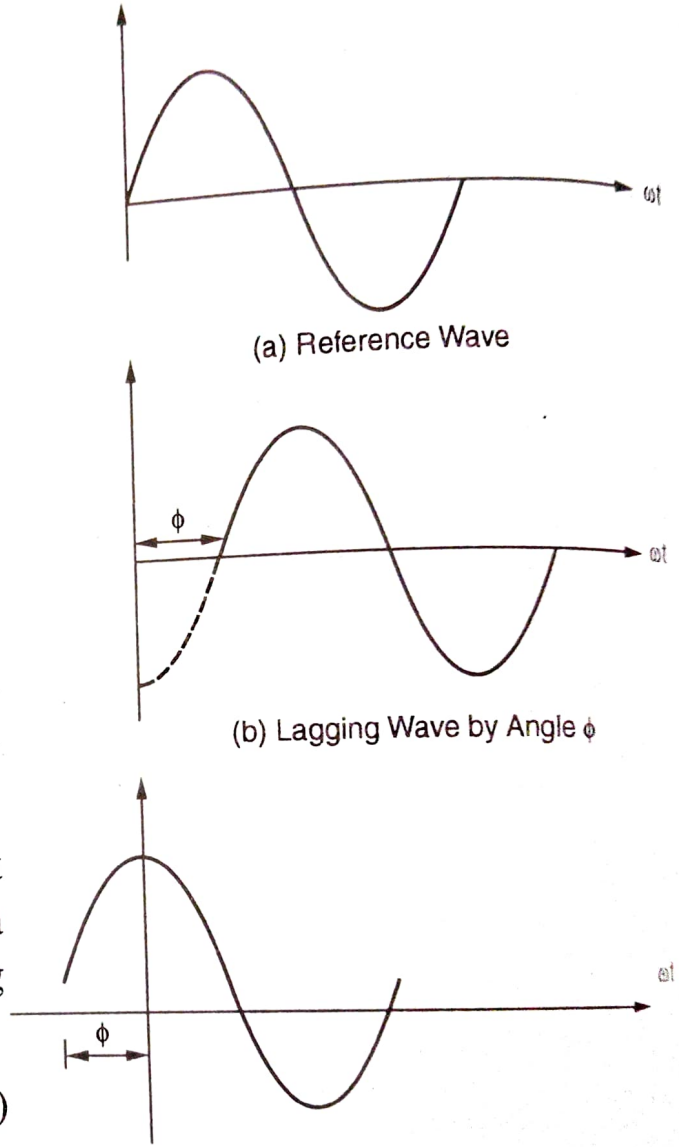
Phase. Phase of any alternating quantity shows the position of wave at any time after it has passed through the zero position of reference.

Phase angle is the angle of alternating quantity with respect to the reference position.

Phase difference. When the maxima and minima of two sinusoidal alternating quantities (of same frequency) are not occurred at the same instant of time, then these two quantities are said to have phase difference.

Lagging current by angle ϕ. If the current wave form of Fig. 4.2 is assumed to be a reference wave then Fig. 4.4 shows the lagging current.

$$i = I_m \sin(\omega t - \phi) \qquad ...(4.4)$$



(a) Reference Wave

(b) Lagging Wave by Angle ϕ

(c) Leading Wave by Angle ϕ

Fig. 4.3.

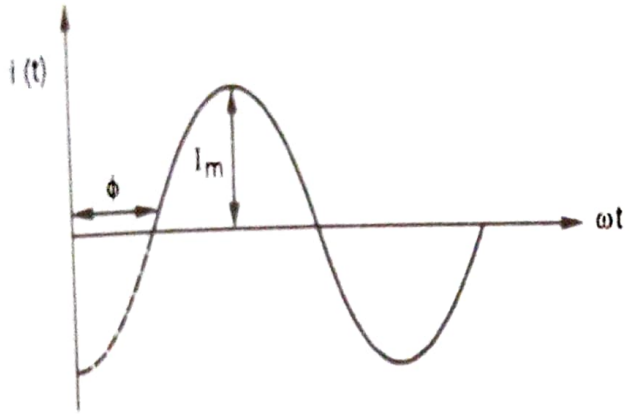


Fig. 4.4. Lagging current.

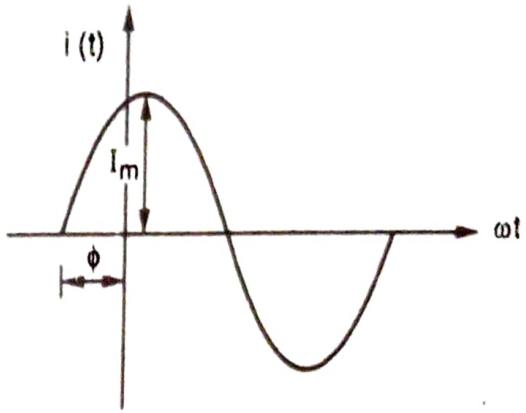


Fig. 4.5. Leading current.

This equation shows that current is lagging (or running behind) the reference wave by an angle ϕ.

Leading current by an angle ϕ. Fig. 4.5 shows the leading current by an angle ϕ by the reference wave of Fig. 4.2.

$$i = I_m \sin(\omega t + \phi). \qquad ...(4.5)$$

4.3. PHASOR DIAGRAM

Phasor diagram or vector diagram shows the graphical representation of alternating quantities in space and shows the phase relation between these quantities. In phasor diagram RMS values (Root mean square values) are represented with their angles. All phasors represent the sine wave of same frequency.

Figs. 4.6, 4.7, 4.8 show the phasor diagrams of voltage and current phasors. Horizontal line is taken as reference. Leading phasors are drawn with angle measured in anticlockwise direction with the reference line while the lagging phasors are drawn with angle measured in clockwise direction with reference line.

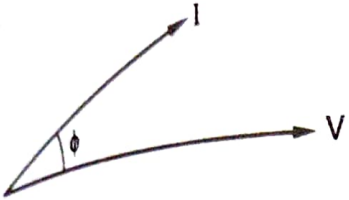


Fig. 4.6.

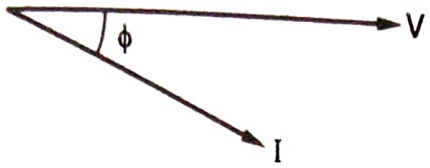


Fig. 4.7.

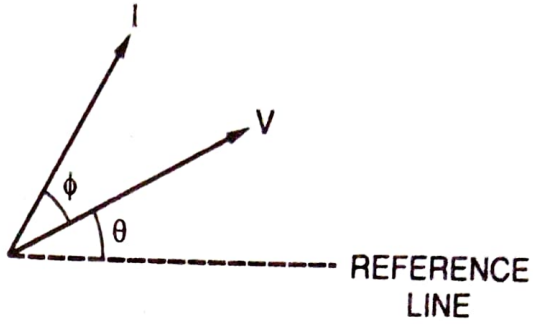


Fig. 4.8.

(*i*) Voltage is reference and current is leading by an angle ϕ, (Fig. 4.6)

$$\bar{V} = |V| \angle 0^\circ$$

$$\bar{I} = |I| \angle \phi$$

(*ii*) Voltage in reference and current is lagging by an angle ϕ, (Fig. 4.7)

$$V = |V| \angle 0^\circ$$

$$I = |I| \angle -\phi$$

(*iii*) Voltage is not reference but current is leading by an angle ϕ, (Fig. 4.8)

$$V = |V| \angle \theta$$

$$I = |I| \angle \theta + \phi$$

Polar representation. Any alternating quantity can be represented in Polar form (magnitude-angle form) as under.

$$i = r_1 \angle \theta_1$$

$$v = r_2 \angle \theta_2 \qquad ...(4.7)$$

Rectangular to Polar conversion :

$$r_1 = \sqrt{a^2 + b^2} \qquad ...(4.8)$$

and $\theta_1 = \tan^{-1} b/a$

Also $r_2 = \sqrt{c^2 + d^2}$

and $\theta_2 = \tan^{-1} d/c \qquad ...(4.9)$

In this way rectangular quantities $a + jb$ and $c + jd$ can be converted into corresponding polar form *i.e.*, $r_1 \angle \theta_1$ and $r_2 \angle \theta_2$.

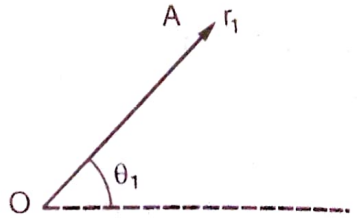


Fig. 4.10.

In Fig. 4.10, $r_1 = OA = \sqrt{a^2 + b^2}$

and θ_1 is the angle made of line OA from the horizontal axis are shown.

Polar to rectangular conversion. $r_1 \angle \theta_1$ can be written as :

$r_1 e^{j\theta_1}$, then by Eular's theorem :

$$r_1 e^{j\theta_1} = r_1 [\cos \theta_1 + j \sin \theta_1] = r_1 \cos \theta_1 + j r_1 \sin \theta_1 \qquad ...(4.10)$$

So, $a = r_1 \cos \theta_1$

and $b = r_1 \sin \theta_1$.

Imag... written as $a \angle -90°$.

4.5. RMS AND AVERAGE VALUE

These two values represent the direct values (steady values) of any alternating quantity and defined as :

4.5.1. RMS value (Root mean square value). The root mean square current is the equivalent steady current of any alternating current which will produce the same amount of heat in a given circuit as could be produced by the alternating current for the same time period.

Consider the circuit shown in Fig. 4.11. First we connect the Direct source (D.C.) to the resistor for the particular time t and find the heat i^2Rt joules. Then for the same time t, we apply the alternating current from A.C. source v and again calculate the heat transferred to the resistor. Now the RMS value of current i is given by,

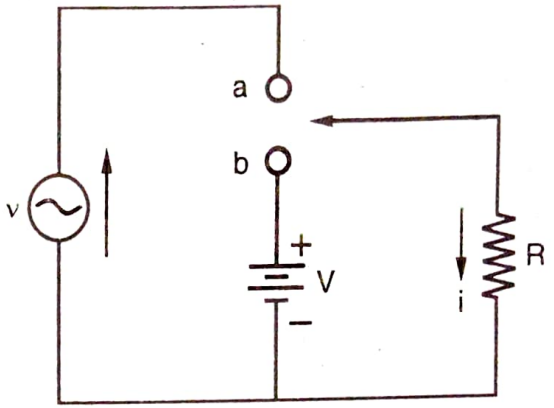


Fig. 4.11.

$$I_{rms} = \sqrt{\frac{1}{T}\int_0^T [i\,(t)]^2\,dt} \qquad ...(4.12)$$

T = Time period of current wave.

RMS value is also known as the effective or virtual value.

4.5.2. RMS Value of Complex Wave. Let the complex current is given by, $i = I_0 + I_{m1}\cos\omega t + I_{m2}\cos 2\omega t + I_{m3}\sin(\omega t - \pi/2)$.

Then the RMS value is given by

$$I_{rms} = \sqrt{I_0^2 + \left(\frac{I_{m1}}{\sqrt{2}}\right)^2 + \left(\frac{I_{m2}}{\sqrt{2}}\right)^2 + \left(\frac{I_{m3}}{\sqrt{2}}\right)^2} \qquad ...(4.13)$$

4.5.3. Average value. The average value of alternating current i is defined by that

steady value of current which transfers the same amount of charge to given circuit as could be transferred by the alternating current i for the same time.

$$I_{av} = \frac{1}{T} \int_0^T i\,(t)\,dt. \qquad \ldots(4.14)$$

Note. Waveforms with half wave symmetry *i.e.*,

$$f\,(t) = -f\left(t + \frac{T}{2}\right)$$

where $T =$ time period

Then the average value of such waves for the complete time period is zero (because positive and negative halves are equal in one time period). For such waves, the average value is computed over the positive half of the period. This is sometimes called the *half cycle average*. (All the waveforms of Fig. 4.1, have such symmetry).

4.5.4. RMS Value of Sinusoidal Wave. Consider the waveform of Fig. 4.2.

The time period is 2π radian *i.e.*, $T = 2\pi$

from equation 4.12 the *RMS value* is given by

$$I_{rms} = \sqrt{\frac{1}{T}\int_0^T [i\,(t)]^2\,dt}$$

Here we integrate with respect to ωt or angle.

So, $$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} [i\,(t)]^2\; d\,(\omega t)}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi} I_m^2 \sin^2 \omega t\, d\,(\omega t)}$$

$$= \sqrt{\frac{I_m^{\,2}}{2\pi}\int_0^{2\pi} \left[\frac{1-\cos 2\omega t}{2}\right] d\,(\omega t)} \qquad \left[\because \quad \sin^2 \omega t = \frac{1-\cos 2\omega t}{2}\right]$$

$$= \sqrt{\frac{I_m^{\,2}}{4\pi}\left[\omega t - \frac{\sin 2\omega t}{2}\right]_0^{2\pi}} = \frac{I_m}{\sqrt{2}} = 0.707\, I_m\,. \qquad \ldots(4.15)$$

4.5.5. Average Value of Sinusoidal Wave. Since the waveform has the half wave symmetry the average value for time period $T = 2\pi$ will be zero. So the average value will be calculated for half time period.

From equation 4.14, the average value is given by

$$I_{av} = \frac{1}{T}\int_0^T i\,(t)\,dt = \frac{1}{\pi}\int_0^{\pi} I_m \sin \omega t\, d\,(\omega t)$$

$$= \frac{I_m}{\pi}\,[-\cos \omega t\,]_0^{\pi} = \frac{2\,I_m}{\pi} = \frac{I_m}{\pi/2} = 0.637\, I_m\,. \qquad \ldots(4.16)$$

4.5.6. Form Factor. The form factor is defined as

$$K_f = \frac{\text{RMS value}}{\text{Average value}} \qquad ...(4.17)$$

For sinusoidal wave,

$$K_f = \frac{0.707\, I_m}{0.637\, I_m} = 1.1 \cdot \qquad ...(4.18)$$

4.5.7. Peak, Crest or Amplitude Factor :

$$K_p = \frac{\text{Maximum value}}{\text{Rms value}} \qquad ...(4.19)$$

For sinusoidal wave, $K_p = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2} = 1.414$ ...(4.20)

4.6. RESPONSE OF SINUSOIDAL ALTERNATING VOLTAGE IN RESISTIVE, INDUCTIVE AND CAPACITIVE CIRCUITS

Before the derivation we define some important terms related to A.C. circuit

4.6.1. Important terms :

(*i*) **Reactance.** That is the opposition in ohm, offered by inductance and capacitance in the circuit. It is given by letter '*X*'.

Inductive reactance : $X_L = \omega L$ in ohms ...(4.21)

where ω = angular frequency in rad/sec.

$= 2\pi f$

f is the frequency in Hz or cycle/sec.

So, $X_L = 2\pi f L$ ...(4.22)

Capacitive reactance : $X_c = \dfrac{1}{\omega C}$ in ohm ...(4.23)

or $X_c = \dfrac{1}{2\pi f C}$ ...(4.24)

(*ii*) **Impedance.** It is the complex summation of resistance and reactance. Its unit is ohm and it is given by letter 'Z'.

$$Z = R + jX \text{ in ohm} \quad ...(4.25)$$

$$= R + j(X_L - X_C) = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

Magnitude of Z, $|Z| = \sqrt{R^2 + X^2}$

Angle of Z $\angle Z = \phi = \tan^{-1}\dfrac{X}{R}$

This is also the angle between voltage and current in the circuit and known as the power factor angle.

(*iii*) **Impedance triangle.** It is the right angle triangle shown in Fig. 4.12. Angle ϕ is the power factor angle.

$$|Z| = \sqrt{R^2 + X^2}$$

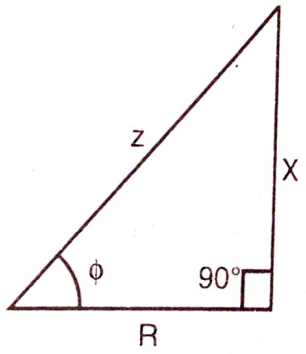


Fig. 4.12. Impedance triangle.

(*iv*) **Admittance.** Reciprocal of impedance is known as the admittance. It is denoted by '*Y*' and its unit is siemens or mho.

$$Y = \frac{1}{Z} = \frac{I}{V}$$

Also,
$$Y = G + jB \qquad ...(4.26)$$

where G = Conductance

and B = Susceptance

The capacitive susceptance is taken as positive while the inductive susceptane is negative.

negative.

(v) Power. There are three types of powers in alternating circuit.

(a) Active power. It is the power which is actually consumed in the circuit (or in resistor). It's unit is watt and is denoted by P. It is also known as the real power.

$$P = VI \cos \phi \text{ watts}$$
$$= I^2 R \qquad \ldots(4.27)$$

(b) Reactive Power. It is the power taken by the reactance of the circuit. This power is not actually consumed but it flows forth and back between source and load. It's unit is Volt-Amp-reactive (VAr). It is given by

$$Q = VI \sin \theta \text{ in VAr}$$
$$= I^2 X \quad \text{where } X = \text{reactance in } \Omega \qquad \ldots(4.28)$$

(c) Apparent Power. It is given by the product of r.m.s. values of voltage and current. Its unit is volt-amp and is given by

$$S = VI \quad \text{in volt–Amp}$$
$$= I^2 Z \qquad \ldots(4.29)$$

where Z is the impedance in Ω.

These three powers may be represented in complex form

$$S = P + jQ \qquad \ldots(4.30)$$

(vi) Power triangle. All the three powers S, P and Q can be represented in a triangle, called the power triangle, shown in Fig. 4.13.

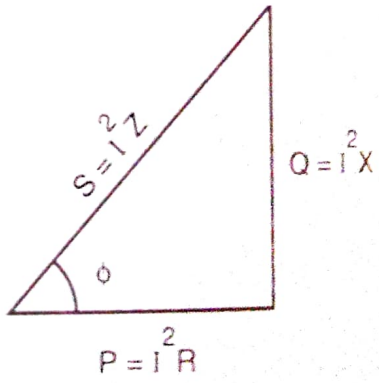


Fig. 4.13. Power triangle.

$$S = \sqrt{P^2 + Q^2} \text{ and } \phi = \tan^{-1}\frac{Q}{P}$$

(*vii*) **Power Factor.** It is the cosine of the angle between voltage and current. Also it is the ratio of the resistance (R) and impedance (Z).

$$\text{P.F.} = \cos\phi \qquad ...(4.31)$$

or from Fig. 4.12,

$$\cos\phi = \frac{R}{Z} \qquad ...(4.32)$$

From Fig. 4.13,

$$\cos\phi = \frac{P}{S} = \frac{\text{Active power}}{\text{Apparent power}} \qquad ...(4.33)$$

P.f. is lagging for inductive circuit and leading for capacitive circuit.

(*viii*) **Quality factor (Q-factor) of the coil.** Q-factor is the reciprocal of the power factor of the coil. It is the figure of merit *i.e.* it should be as large as possible for any coil (combination of resistance and inductance)

$$Q = 2\pi \frac{\text{Maximum energy stored}}{\text{Energy dissipated per cycle}} \qquad ...(4.34)$$

or

$$Q = \frac{1}{\cos\phi} = \frac{Z}{R} = \frac{\sqrt{R^2 + X_L^2}}{R} \qquad ...(4.35)$$

If R is very less as compared to X_L then,

$$Q = \frac{X_L}{R} = \frac{\omega L}{R} \qquad ...(4.36)$$